

8.1 How to read this chapter

[Table 75](#) shows the functions of the PINSEL registers in the LPC176x/5x.

Table 75. Summary of PINSEL registers

Register	Controls	Table
PINSEL0	P0[15:0]	Table 80
PINSEL1	P0 [31:16]	Table 81
PINSEL2	P1 [15:0] (Ethernet)	Table 82
PINSEL3	P1 [31:16]	Table 83
PINSEL4	P2 [15:0]	Table 84
PINSEL5	P2 [31:16]	not used
PINSEL6	P3 [15:0]	not used
PINSEL7	P3 [31:16]	Table 85
PINSEL8	P4 [15:0]	not used
PINSEL9	P4 [31:16]	Table 86
PINSEL10	Trace port enable	Table 87

8.2 Description

The pin connect block allows most pins of the microcontroller to have more than one potential function. Configuration registers control the multiplexers to allow connection between the pin and the on chip peripherals.

Peripherals should be connected to the appropriate pins prior to being activated and prior to any related interrupt(s) being enabled. Activity of any enabled peripheral function that is not mapped to a related pin should be considered undefined.

Selection of a single function on a port pin excludes other peripheral functions available on the same pin. However, the GPIO input stays connected and may be read by software or used to contribute to the GPIO interrupt feature.

8.3 Pin function select register values

The PINSEL registers control the functions of device pins as shown below. Pairs of bits in these registers correspond to specific device pins.

Table 76. Pin function select register bits

PINSEL0 to PINSEL9 Values	Function	Value after Reset
00	Primary (default) function, typically GPIO port	00
01	First alternate function	
10	Second alternate function	
11	Third alternate function	

The direction control bit in the GPIO registers is effective only when the GPIO function is selected for a pin. For other functions, direction is controlled automatically. Each derivative typically has a different pinout and therefore a different set of functions possible for each pin. Details for a specific derivative may be found in the appropriate data sheet.

Multiple connections

Since a particular peripheral function may be allowed on more than one pin, it is in principle possible to configure more than one pin to perform the same function. If a peripheral output function is configured to appear on more than one pin, it will in fact be routed to those pins. If a peripheral input function is configured to appear on more than one pin for some reason, the peripheral will receive its input from the lowest port number. For instance, any pin of port 0 will take precedence over any pin of a higher numbered port, and pin 0 of any port will take precedence over a higher numbered pin of the same port.

8.4 Pin mode select register values

The PINMODE registers control the input mode of all ports. This includes the use of the on-chip pull-up/pull-down resistor feature and a special open drain operating mode. The on-chip pull-up/pull-down resistor can be selected for every port pin regardless of the function on this pin with the exception of the I²C pins for the I2C0 interface and the USB pins (see [Section 8.5.10](#)). Three bits are used to control the mode of a port pin, two in a PINMODE register, and an additional one in a PINMODE_OD register. Bits are reserved for unused pins as in the PINSEL registers.

Table 77. Pin Mode Select register Bits

PINMODE0 to PINMODE9 Values	Function	Value after Reset
00	Pin has an on-chip pull-up resistor enabled.	00
01	Repeater mode (see text below).	
10	Pin has neither pull-up nor pull-down resistor enabled.	
11	Pin has an on-chip pull-down resistor enabled.	

Repeater mode enables the pull-up resistor if the pin is at a logic high and enables the pull-down resistor if the pin is at a logic low. This causes the pin to retain its last known state if it is configured as an input and is not driven externally. The state retention is not applicable to the Deep Power-down mode. Repeater mode may typically be used to prevent a pin from floating (and potentially using significant power if it floats to an indeterminate state) if it is temporarily not driven.

The PINMODE_OD registers control the open drain mode for ports. The open drain mode causes the pin to be pulled low normally if it is configured as an output and the data value is 0. If the data value is 1, the output drive of the pin is turned off, equivalent to changing the pin direction. This combination simulates an open drain output.

Table 78. Open Drain Pin Mode Select register Bits

PINMODE_OD0 to PINMODE_OD4 Values	Function	Value after Reset
0	Pin is in the normal (not open drain) mode.	00
1	Pin is in the open drain mode.	

Function of PINMODE in open drain mode

Normally the value of PINMODE applies to a pin only when it is in the input mode. When a pin is in the open drain mode, caused by a 1 in the corresponding bit of one of the PINMODE_OD registers, the input mode still does not apply when the pin is outputting a 0. However, when the pin value is 1, PINMODE applies since this state turns off the pin's output driver. For example, this allows for the possibility of configuring a pin to be open drain with an on-chip pullup. A pullup in this case which is only on when the pin is not being pulled low by the pin's own output.

8.5 Register description

The Pin Control Module contains 11 registers as shown in [Table 79](#) below.

Table 79. Pin Connect Block Register Map

Name	Description	Access	Reset Value ^[1]	Address
PINSEL0	Pin function select register 0.	R/W	0	0x4002 C000
PINSEL1	Pin function select register 1.	R/W	0	0x4002 C004
PINSEL2	Pin function select register 2.	R/W	0	0x4002 C008
PINSEL3	Pin function select register 3.	R/W	0	0x4002 C00C
PINSEL4	Pin function select register 4	R/W	0	0x4002 C010
PINSEL7	Pin function select register 7	R/W	0	0x4002 C01C
PINSEL8	Pin function select register 8	R/W	0	0x4002 C020
PINSEL9	Pin function select register 9	R/W	0	0x4002 C024
PINSEL10	Pin function select register 10	R/W	0	0x4002 C028
PINMODE0	Pin mode select register 0	R/W	0	0x4002 C040
PINMODE1	Pin mode select register 1	R/W	0	0x4002 C044
PINMODE2	Pin mode select register 2	R/W	0	0x4002 C048
PINMODE3	Pin mode select register 3.	R/W	0	0x4002 C04C
PINMODE4	Pin mode select register 4	R/W	0	0x4002 C050
PINMODE5	Pin mode select register 5	R/W	0	0x4002 C054
PINMODE6	Pin mode select register 6	R/W	0	0x4002 C058
PINMODE7	Pin mode select register 7	R/W	0	0x4002 C05C
PINMODE9	Pin mode select register 9	R/W	0	0x4002 C064
PINMODE_OD0	Open drain mode control register 0	R/W	0	0x4002 C068
PINMODE_OD1	Open drain mode control register 1	R/W	0	0x4002 C06C
PINMODE_OD2	Open drain mode control register 2	R/W	0	0x4002 C070
PINMODE_OD3	Open drain mode control register 3	R/W	0	0x4002 C074
PINMODE_OD4	Open drain mode control register 4	R/W	0	0x4002 C078
I2CPADCFG	I ² C Pin Configuration register	R/W	0	0x4002 C07C

[1] Reset Value reflects the data stored in used bits only. It does not include reserved bits content.

Pin control module register reset values

On external reset, watchdog reset, power-on-reset (POR), and BOD reset, all registers in this module are reset to '0'.

8.5.1 Pin Function Select register 0 (PINSEL0 - 0x4002 C000)

The PINSEL0 register controls the functions of the lower half of Port 0. The direction control bit in FIODIR register is effective only when the GPIO function is selected for a pin. For other functions, the direction is controlled automatically.

Table 80. Pin function select register 0 (PINSEL0 - address 0x4002 C000) bit description

PINSEL0	Pin name	Function when 00	Function when 01	Function when 10	Function when 11	Reset value
1:0	P0.0	GPIO Port 0.0	RD1	TXD3	SDA1	00
3:2	P0.1	GPIO Port 0.1	TD1	RXD3	SCL1	00
5:4	P0.2	GPIO Port 0.2	TXD0	AD0.7	Reserved	00
7:6	P0.3	GPIO Port 0.3	RXD0	AD0.6	Reserved	00
9:8	P0.4 ^[1]	GPIO Port 0.4	I2SRX_CLK	RD2	CAP2.0	00
11:10	P0.5 ^[1]	GPIO Port 0.5	I2SRX_WS	TD2	CAP2.1	00
13:12	P0.6	GPIO Port 0.6	I2SRX_SDA	SSEL1	MAT2.0	00
15:14	P0.7	GPIO Port 0.7	I2STX_CLK	SCK1	MAT2.1	00
17:16	P0.8	GPIO Port 0.8	I2STX_WS	MISO1	MAT2.2	00
19:18	P0.9	GPIO Port 0.9	I2STX_SDA	MOSI1	MAT2.3	00
21:20	P0.10	GPIO Port 0.10	TXD2	SDA2	MAT3.0	00
23:22	P0.11	GPIO Port 0.11	RXD2	SCL2	MAT3.1	00
29:24	-	Reserved	Reserved	Reserved	Reserved	0
31:30	P0.15	GPIO Port 0.15	TXD1	SCK0	SCK	00

[1] Not available on 80-pin package.

8.5.2 Pin Function Select Register 1 (PINSEL1 - 0x4002 C004)

The PINSEL1 register controls the functions of the upper half of Port 0. The direction control bit in the FIODIR register is effective only when the GPIO function is selected for a pin. For other functions the direction is controlled automatically.

Table 81. Pin function select register 1 (PINSEL1 - address 0x4002 C004) bit description

PINSEL1	Pin name	Function when 00	Function when 01	Function when 10	Function when 11	Reset value
1:0	P0.16	GPIO Port 0.16	RXD1	SSEL0	SSEL	00
3:2	P0.17	GPIO Port 0.17	CTS1	MISO0	MISO	00
5:4	P0.18	GPIO Port 0.18	DCD1	MOSI0	MOSI	00
7:6	P0.19 ^[1]	GPIO Port 0.19	DSR1	Reserved	SDA1	00
9:8	P0.20 ^[1]	GPIO Port 0.20	DTR1	Reserved	SCL1	00
11:10	P0.21 ^[1]	GPIO Port 0.21	RI1	Reserved	RD1	00
13:12	P0.22	GPIO Port 0.22	RTS1	Reserved	TD1	00
15:14	P0.23 ^[1]	GPIO Port 0.23	AD0.0	I2SRX_CLK	CAP3.0	00
17:16	P0.24 ^[1]	GPIO Port 0.24	AD0.1	I2SRX_WS	CAP3.1	00
19:18	P0.25	GPIO Port 0.25	AD0.2	I2SRX_SDA	TXD3	00
21:20	P0.26	GPIO Port 0.26	AD0.3	AOUT	RXD3	00
23:22	P0.27 ^{[1][2]}	GPIO Port 0.27	SDA0	USB_SDA	Reserved	00

Table 81. Pin function select register 1 (PINSEL1 - address 0x4002 C004) bit description ...continued

PINSEL1	Pin name	Function when 00	Function when 01	Function when 10	Function when 11	Reset value
25:24	P0.28 ^{[1][2]}	GPIO Port 0.28	SCL0	USB_SCL	Reserved	00
27:26	P0.29	GPIO Port 0.29	USB_D+	Reserved	Reserved	00
29:28	P0.30	GPIO Port 0.30	USB_D-	Reserved	Reserved	00
31:30	-	Reserved	Reserved	Reserved	Reserved	00

[1] Not available on 80-pin package.

[2] Pins P0[27] and P0[28] are open-drain for I²C-bus compliance.

8.5.3 Pin Function Select register 2 (PINSEL2 - 0x4002 C008)

The PINSEL2 register controls the functions of the lower half of Port 1, which contains the Ethernet related pins. The direction control bit in the FIO1DIR register is effective only when the GPIO function is selected for a pin. For other functions, the direction is controlled automatically.

Table 82. Pin function select register 2 (PINSEL2 - address 0x4002 C008) bit description

PINSEL2	Pin name	Function when 00	Function when 01	Function when 10	Function when 11	Reset value
1:0	P1.0	GPIO Port 1.0	ENET_TXD0	Reserved	Reserved	00
3:2	P1.1	GPIO Port 1.1	ENET_TXD1	Reserved	Reserved	00
7:4	-	Reserved	Reserved	Reserved	Reserved	0
9:8	P1.4	GPIO Port 1.4	ENET_TX_EN	Reserved	Reserved	00
15:10	-	Reserved	Reserved	Reserved	Reserved	0
17:16	P1.8	GPIO Port 1.8	ENET_CRS	Reserved	Reserved	00
19:18	P1.9	GPIO Port 1.9	ENET_RXD0	Reserved	Reserved	00
21:20	P1.10	GPIO Port 1.10	ENET_RXD1	Reserved	Reserved	00
27:22	-	Reserved	Reserved	Reserved	Reserved	0
29:28	P1.14	GPIO Port 1.14	ENET_RX_ER	Reserved	Reserved	00
31:30	P1.15	GPIO Port 1.15	ENET_REF_CLK	Reserved	Reserved	00

8.5.4 Pin Function Select Register 3 (PINSEL3 - 0x4002 C00C)

The PINSEL3 register controls the functions of the upper half of Port 1. The direction control bit in the FIO1DIR register is effective only when the GPIO function is selected for a pin. For other functions, direction is controlled automatically.

Table 83. Pin function select register 3 (PINSEL3 - address 0x4002 C00C) bit description

PINSEL3	Pin name	Function when 00	Function when 01	Function when 10	Function when 11	Reset value
1:0	P1.16 ^[1]	GPIO Port 1.16	ENET_MDC	Reserved	Reserved	00
3:2	P1.17 ^[1]	GPIO Port 1.17	ENET_MDIO	Reserved	Reserved	00
5:4	P1.18	GPIO Port 1.18	USB_UP_LED	PWM1.1	CAP1.0	00
7:6	P1.19	GPIO Port 1.19	MCOA0	USB_PPWR	CAP1.1	00
9:8	P1.20	GPIO Port 1.20	MCI0	PWM1.2	SCK0	00

Table 83. Pin function select register 3 (PINSEL3 - address 0x4002 C00C) bit description ...continued

PINSEL3	Pin name	Function when 00	Function when 01	Function when 10	Function when 11	Reset value
11:10	P1.21 ^[1]	GPIO Port 1.21	MCABORT	PWM1.3	SSEL0	00
13:12	P1.22	GPIO Port 1.22	MCOB0	USB_PWRD	MAT1.0	00
15:14	P1.23	GPIO Port 1.23	MCI1	PWM1.4	MISO0	00
17:16	P1.24	GPIO Port 1.24	MCI2	PWM1.5	MOSI0	00
19:18	P1.25	GPIO Port 1.25	MCOA1	Reserved	MAT1.1	00
21:20	P1.26	GPIO Port 1.26	MCOB1	PWM1.6	CAP0.0	00
23:22	P1.27 ^[1]	GPIO Port 1.27	CLKOUT	USB_OVRCCR	CAP0.1	00
25:24	P1.28	GPIO Port 1.28	MCOA2	PCAP1.0	MAT0.0	00
27:26	P1.29	GPIO Port 1.29	MCOB2	PCAP1.1	MAT0.1	00
29:28	P1.30	GPIO Port 1.30	Reserved	V _{BUS}	AD0.4	00
31:30	P1.31	GPIO Port 1.31	Reserved	SCK1	AD0.5	00

[1] Not available on 80-pin package.

8.5.5 Pin Function Select Register 4 (PINSEL4 - 0x4002 C010)

The PINSEL4 register controls the functions of the lower half of Port 2. The direction control bit in the FIO2DIR register is effective only when the GPIO function is selected for a pin. For other functions, direction is controlled automatically.

Table 84. Pin function select register 4 (PINSEL4 - address 0x4002 C010) bit description

PINSEL4	Pin name	Function when 00	Function when 01	Function when 10	Function when 11	Reset value
1:0	P2.0	GPIO Port 2.0	PWM1.1	TXD1	Reserved	00
3:2	P2.1	GPIO Port 2.1	PWM1.2	RXD1	Reserved	00
5:4	P2.2	GPIO Port 2.2	PWM1.3	CTS1	Reserved ^[2]	00
7:6	P2.3	GPIO Port 2.3	PWM1.4	DCD1	Reserved ^[2]	00
9:8	P2.4	GPIO Port 2.4	PWM1.5	DSR1	Reserved ^[2]	00
11:10	P2.5	GPIO Port 2.5	PWM1.6	DTR1	Reserved ^[2]	00
13:12	P2.6	GPIO Port 2.6	PCAP1.0	RI1	Reserved ^[2]	00
15:14	P2.7	GPIO Port 2.7	RD2	RTS1	Reserved	00
17:16	P2.8	GPIO Port 2.8	TD2	TXD2	ENET_MDC	00
19:18	P2.9	GPIO Port 2.9	USB_CONNECT	RXD2	ENET_MDIO	00
21:20	P2.10	GPIO Port 2.10	EINT0	NMI	Reserved	00
23:22	P2.11 ^[1]	GPIO Port 2.11	EINT1	Reserved	I2STX_CLK	00
25:24	P2.12 ^[1]	GPIO Port 2.12	EINT2	Reserved	I2STX_WS	00
27:26	P2.13 ^[1]	GPIO Port 2.13	EINT3	Reserved	I2STX_SDA	00
31:28	-	Reserved	Reserved	Reserved	Reserved	0

[1] Not available on 80-pin package.

[2] These pins support a debug trace function when selected via a development tool or by writing to the PINSEL10 register. See [Section 8.5.8 "Pin Function Select Register 10 \(PINSEL10 - 0x4002 C028\)"](#) for details.

8.5.6 Pin Function Select Register 7 (PINSEL7 - 0x4002 C01C)

The PINSEL7 register controls the functions of the upper half of Port 3. The direction control bit in the FIO3DIR register is effective only when the GPIO function is selected for a pin. For other functions, direction is controlled automatically.

Table 85. Pin function select register 7 (PINSEL7 - address 0x4002 C01C) bit description

PINSEL7	Pin name	Function when 00	Function when 01	Function when 10	Function when 11	Reset value
17:0	-	Reserved	Reserved	Reserved	Reserved	0
19:18	P3.25 ^[1]	GPIO Port 3.25	Reserved	MAT0.0	PWM1.2	00
21:20	P3.26 ^[1]	GPIO Port 3.26	STCLK	MAT0.1	PWM1.3	00
31:22	-	Reserved	Reserved	Reserved	Reserved	0

[1] Not available on 80-pin package.

8.5.7 Pin Function Select Register 9 (PINSEL9 - 0x4002 C024)

The PINSEL9 register controls the functions of the upper half of Port 4. The direction control bit in the FIO4DIR register is effective only when the GPIO function is selected for a pin. For other functions, direction is controlled automatically.

Table 86. Pin function select register 9 (PINSEL9 - address 0x4002 C024) bit description

PINSEL9	Pin name	Function when 00	Function when 01	Function when 10	Function when 11	Reset value
23:0	-	Reserved	Reserved	Reserved	Reserved	00
25:24	P4.28	GPIO Port 4.28	RX_MCLK	MAT2.0	TXD3	00
27:26	P4.29	GPIO Port 4.29	TX_MCLK	MAT2.1	RXD3	00
31:28	-	Reserved	Reserved	Reserved	Reserved	00

8.5.8 Pin Function Select Register 10 (PINSEL10 - 0x4002 C028)

Only bit 3 of this register is used to control the Trace function on pins P2.2 through P2.6.

Table 87. Pin function select register 10 (PINSEL10 - address 0x4002 C028) bit description

Bit	Symbol	Value	Description	Reset value
2:0	-	-	Reserved. Software should not write 1 to these bits.	NA
3	GPIO/TRACE		TPIU interface pins control.	0
		0	TPIU interface is disabled.	
		1	TPIU interface is enabled. TPIU signals are available on the pins hosting them regardless of the PINSEL4 content.	
31:4	-	-	Reserved. Software should not write 1 to these bits.	NA

8.5.9 Pin Mode select register 0 (PINMODE0 - 0x4002 C040)

This register controls pull-up/pull-down resistor configuration for Port 0 pins 0 to 15.

Table 88. Pin Mode select register 0 (PINMODE0 - address 0x4002 C040) bit description

PINMODE0	Symbol	Value	Description	Reset value
1:0	P0.00MODE		Port 0 pin 0 on-chip pull-up/down resistor control.	00
		00	P0.0 pin has a pull-up resistor enabled.	
		01	P0.0 pin has repeater mode enabled.	
		10	P0.0 pin has neither pull-up nor pull-down.	
		11	P0.0 has a pull-down resistor enabled.	
3:2	P0.01MODE		Port 0 pin 1 control, see P0.00MODE.	00
5:4	P0.02MODE		Port 0 pin 2 control, see P0.00MODE.	00
7:6	P0.03MODE		Port 0 pin 3 control, see P0.00MODE.	00
9:8	P0.04MODE ^[1]		Port 0 pin 4 control, see P0.00MODE.	00
11:10	P0.05MODE ^[1]		Port 0 pin 5 control, see P0.00MODE.	00
13:12	P0.06MODE		Port 0 pin 6 control, see P0.00MODE.	00
15:14	P0.07MODE		Port 0 pin 7 control, see P0.00MODE.	00
17:16	P0.08MODE		Port 0 pin 8 control, see P0.00MODE.	00
19:18	P0.09MODE		Port 0 pin 9 control, see P0.00MODE.	00
21:20	P0.10MODE		Port 0 pin 10 control, see P0.00MODE.	00
23:22	P0.11MODE		Port 0 pin 11 control, see P0.00MODE.	00
29:24	-		Reserved.	NA
31:30	P0.15MODE		Port 0 pin 15 control, see P0.00MODE.	00

[1] Not available on 80-pin package.

8.5.10 Pin Mode select register 1 (PINMODE1 - 0x4002 C044)

This register controls pull-up/pull-down resistor configuration for Port 1 pins 16 to 26. For details see [Section 8.4 “Pin mode select register values”](#).

Table 89. Pin Mode select register 1 (PINMODE1 - address 0x4002 C044) bit description

PINMODE1	Symbol	Description	Reset value
1:0	P0.16MODE	Port 0 pin 16 control, see P0.00MODE.	00
3:2	P0.17MODE	Port 0 pin 17 control, see P0.00MODE.	00
5:4	P0.18MODE	Port 0 pin 18 control, see P0.00MODE.	00
7:6	P0.19MODE ^[1]	Port 0 pin 19 control, see P0.00MODE.	00
9:8	P0.20MODE ^[1]	Port 0 pin 20 control, see P0.00MODE.	00
11:10	P0.21MODE ^[1]	Port 0 pin 21 control, see P0.00MODE.	00
13:12	P0.22MODE	Port 0 pin 22 control, see P0.00MODE.	00
15:14	P0.23MODE ^[1]	Port 0 pin 23 control, see P0.00MODE.	00
17:16	P0.24MODE ^[1]	Port 0 pin 24 control, see P0.00MODE.	00
19:18	P0.25MODE	Port 0 pin 25 control, see P0.00MODE.	00

Table 89. Pin Mode select register 1 (PINMODE1 - address 0x4002 C044) bit description

PINMODE1	Symbol	Description	Reset value
21:20	P0.26MODE	Port 0 pin 26 control, see P0.00MODE.	00
29:22	-	Reserved. [2]	NA
31:30	-	Reserved.	NA

[1] Not available on 80-pin package.

[2] The pin mode cannot be selected for pins P0[27] to P0[30]. Pins P0[27] and P0[28] are dedicated I2C open-drain pins without pull-up/down. Pins P0[29] and P0[30] are USB specific pins without configurable pull-up or pull-down resistors. Pins P0[29] and P0[30] also must have the same direction since they operate as a unit for the USB function, see [Section 9.5.1 "GPIO port Direction register FIOxDIR \(FIO0DIR to FIO4DIR- 0x2009 C000 to 0x2009 C080\)"](#).

8.5.11 Pin Mode select register 2 (PINMODE2 - 0x4002 C048)

This register controls pull-up/pull-down resistor configuration for Port 1 pins 0 to 15. For details see [Section 8.4 "Pin mode select register values"](#).

Table 90. Pin Mode select register 2 (PINMODE2 - address 0x4002 C048) bit description

PINMODE2	Symbol	Description	Reset value
1:0	P1.00MODE	Port 1 pin 0 control, see P0.00MODE.	00
3:2	P1.01MODE	Port 1 pin 1 control, see P0.00MODE.	00
7:4	-	Reserved.	NA
9:8	P1.04MODE	Port 1 pin 4 control, see P0.00MODE.	00
15:10	-	Reserved.	NA
17:16	P1.08MODE	Port 1 pin 8 control, see P0.00MODE.	00
19:18	P1.09MODE	Port 1 pin 9 control, see P0.00MODE.	00
21:20	P1.10MODE	Port 1 pin 10 control, see P0.00MODE.	00
27:22	-	Reserved.	NA
29:28	P1.14MODE	Port 1 pin 14 control, see P0.00MODE.	00
31:30	P1.15MODE	Port 1 pin 15 control, see P0.00MODE.	00

8.5.12 Pin Mode select register 3 (PINMODE3 - 0x4002 C04C)

This register controls pull-up/pull-down resistor configuration for Port 1 pins 16 to 31. For details see [Section 8.4 "Pin mode select register values"](#).

Table 91. Pin Mode select register 3 (PINMODE3 - address 0x4002 C04C) bit description

PINMODE3	Symbol	Description	Reset value
1:0	P1.16MODE[1]	Port 1 pin 16 control, see P0.00MODE.	00
3:2	P1.17MODE[1]	Port 1 pin 17 control, see P0.00MODE.	00
5:4	P1.18MODE	Port 1 pin 18 control, see P0.00MODE.	00
7:6	P1.19MODE	Port 1 pin 19 control, see P0.00MODE.	00
9:8	P1.20MODE	Port 1 pin 20 control, see P0.00MODE.	00
11:10	P1.21MODE[1]	Port 1 pin 21 control, see P0.00MODE.	00
13:12	P1.22MODE	Port 1 pin 22 control, see P0.00MODE.	00

Table 91. Pin Mode select register 3 (PINMODE3 - address 0x4002 C04C) bit description

PINMODE3	Symbol	Description	Reset value
15:14	P1.23MODE	Port 1 pin 23 control, see P0.00MODE.	00
17:16	P1.24MODE	Port 1 pin 24 control, see P0.00MODE.	00
19:18	P1.25MODE	Port 1 pin 25 control, see P0.00MODE.	00
21:20	P1.26MODE	Port 1 pin 26 control, see P0.00MODE.	00
23:22	P1.27MODE ^[1]	Port 1 pin 27 control, see P0.00MODE.	00
25:24	P1.28MODE	Port 1 pin 28 control, see P0.00MODE.	00
27:26	P1.29MODE	Port 1 pin 29 control, see P0.00MODE.	00
29:28	P1.30MODE	Port 1 pin 30 control, see P0.00MODE.	00
31:30	P1.31MODE	Port 1 pin 31 control, see P0.00MODE.	00

[1] Not available on 80-pin package.

8.5.13 Pin Mode select register 4 (PINMODE4 - 0x4002 C050)

This register controls pull-up/pull-down resistor configuration for Port 2 pins 0 to 15. For details see [Section 8.4 “Pin mode select register values”](#).

Table 92. Pin Mode select register 4 (PINMODE4 - address 0x4002 C050) bit description

PINMODE4	Symbol	Description	Reset value
1:0	P2.00MODE	Port 2 pin 0 control, see P0.00MODE.	00
3:2	P2.01MODE	Port 2 pin 1 control, see P0.00MODE.	00
5:4	P2.02MODE	Port 2 pin 2 control, see P0.00MODE.	00
7:6	P2.03MODE	Port 2 pin 3 control, see P0.00MODE.	00
9:8	P2.04MODE	Port 2 pin 4 control, see P0.00MODE.	00
11:10	P2.05MODE	Port 2 pin 5 control, see P0.00MODE.	00
13:12	P2.06MODE	Port 2 pin 6 control, see P0.00MODE.	00
15:14	P2.07MODE	Port 2 pin 7 control, see P0.00MODE.	00
17:16	P2.08MODE	Port 2 pin 8 control, see P0.00MODE.	00
19:18	P2.09MODE	Port 2 pin 9 control, see P0.00MODE.	00
21:20	P2.10MODE	Port 2 pin 10 control, see P0.00MODE.	00
23:22	P2.11MODE ^[1]	Port 2 pin 11 control, see P0.00MODE.	00
25:24	P2.12MODE ^[1]	Port 2 pin 12 control, see P0.00MODE.	00
27:26	P2.13MODE ^[1]	Port 2 pin 13 control, see P0.00MODE.	00
31:28	-	Reserved.	NA

[1] Not available on 80-pin package.

8.5.14 Pin Mode select register 7 (PINMODE7 - 0x4002 C05C)

This register controls pull-up/pull-down resistor configuration for Port 3 pins 16 to 31. For details see [Section 8.4 “Pin mode select register values”](#).

Table 93. Pin Mode select register 7 (PINMODE7 - address 0x4002 C05C) bit description

PINMODE7	Symbol	Description	Reset value
17:0	-	Reserved	NA
19:18	P3.25MODE ^[1]	Port 3 pin 25 control, see P0.00MODE.	00
21:20	P3.26MODE ^[1]	Port 3 pin 26 control, see P0.00MODE.	00
31:22	-	Reserved.	NA

[1] Not available on 80-pin package.

8.5.15 Pin Mode select register 9 (PINMODE9 - 0x4002 C064)

This register controls pull-up/pull-down resistor configuration for Port 4 pins 16 to 31. For details see [Section 8.4 “Pin mode select register values”](#).

Table 94. Pin Mode select register 9 (PINMODE9 - address 0x4002 C064) bit description

PINMODE9	Symbol	Description	Reset value
23:0	-	Reserved.	NA
25:24	P4.28MODE	Port 4 pin 28 control, see P0.00MODE.	00
27:26	P4.29MODE	Port 4 pin 29 control, see P0.00MODE.	00
31:28	-	Reserved.	NA

8.5.16 Open Drain Pin Mode select register 0 (PINMODE_OD0 - 0x4002 C068)

This register controls the open drain mode for Port 0 pins. For details see [Section 8.4 “Pin mode select register values”](#).

Table 95. Open Drain Pin Mode select register 0 (PINMODE_OD0 - address 0x4002 C068) bit description

PINMODE_OD0	Symbol	Value	Description	Reset value
0	P0.00OD ^[3]		Port 0 pin 0 open drain mode control.	0
		0	P0.0 pin is in the normal (not open drain) mode.	
		1	P0.0 pin is in the open drain mode.	
1	P0.01OD ^[3]		Port 0 pin 1 open drain mode control, see P0.00OD	0
2	P0.02OD		Port 0 pin 2 open drain mode control, see P0.00OD	0
3	P0.03OD		Port 0 pin 3 open drain mode control, see P0.00OD	0
4	P0.04OD		Port 0 pin 4 open drain mode control, see P0.00OD	0
5	P0.05OD		Port 0 pin 5 open drain mode control, see P0.00OD	0
6	P0.06OD		Port 0 pin 6 open drain mode control, see P0.00OD	0
7	P0.07OD		Port 0 pin 7 open drain mode control, see P0.00OD	0
8	P0.08OD		Port 0 pin 8 open drain mode control, see P0.00OD	0
9	P0.09OD		Port 0 pin 9 open drain mode control, see P0.00OD	0

Table 95. Open Drain Pin Mode select register 0 (PINMODE_OD0 - address 0x4002 C068) bit description ...continued

PINMODE_OD0	Symbol	Value	Description	Reset value
10	P0.10OD ^[3]		Port 0 pin 10 open drain mode control, see P0.00OD	0
11	P0.11OD ^[3]		Port 0 pin 11 open drain mode control, see P0.00OD	0
14:12	-		Reserved.	NA
15	P0.15OD		Port 0 pin 15 open drain mode control, see P0.00OD	0
16	P0.16OD		Port 0 pin 16 open drain mode control, see P0.00OD	0
17	P0.17OD		Port 0 pin 17 open drain mode control, see P0.00OD	0
18	P0.18OD		Port 0 pin 18 open drain mode control, see P0.00OD	0
19	P0.19OD ^[3]		Port 0 pin 19 open drain mode control, see P0.00OD	0
20	P0.20OD ^[3]		Port 0 pin 20 open drain mode control, see P0.00OD	0
21	P0.21OD		Port 0 pin 21 open drain mode control, see P0.00OD	0
22	P0.22OD		Port 0 pin 22 open drain mode control, see P0.00OD	0
23	P0.23OD		Port 0 pin 23 open drain mode control, see P0.00OD	0
24	P0.24OD		Port 0 pin 24 open drain mode control, see P0.00OD	0
25	P0.25OD		Port 0 pin 25 open drain mode control, see P0.00OD	0
26	P0.26OD		Port 0 pin 26 open drain mode control, see P0.00OD	0
28:27	- ^[2]		Reserved.	NA
29	P0.29OD		Port 0 pin 29 open drain mode control, see P0.00OD	0
30	P0.30OD		Port 0 pin 30 open drain mode control, see P0.00OD	0
31	-		Reserved.	NA

[1] Not available on 80-pin package.

[2] Port 0 pins 27 and 28 should be set up using the I2CPADCFG register if they are used for an I²C-bus. Bits 27 and 28 of PINMODE_OD0 do not have any affect on these pins, they are special open drain I²C-bus compatible pins.

[3] Port 0 bits 1:0, 11:10, and 20:19 may potentially be used for I²C-buses using standard port pins. If so, they should be configured for open drain mode via the related bits in PINMODE_OD0.

8.5.17 Open Drain Pin Mode select register 1 (PINMODE_OD1 - 0x4002 C06C)

This register controls the open drain mode for Port 1 pins. For details see [Section 8.4 "Pin mode select register values"](#).

Table 96. Open Drain Pin Mode select register 1 (PINMODE_OD1 - address 0x4002 C06C) bit description

PINMODE_OD1	Symbol	Value	Description	Reset value
0	P1.00OD		Port 1 pin 0 open drain mode control.	0
		0	P1.0 pin is in the normal (not open drain) mode.	
		1	P1.0 pin is in the open drain mode.	
1	P1.01OD		Port 1 pin 1 open drain mode control, see P1.00OD	0
3:2	-		Reserved.	NA
4	P1.04OD		Port 1 pin 4 open drain mode control, see P1.00OD	0

Table 96. Open Drain Pin Mode select register 1 (PINMODE_OD1 - address 0x4002 C06C) bit description ...continued

PINMODE_OD1	Symbol	Value	Description	Reset value
7:5	-		Reserved.	NA
8	P1.08OD		Port 1 pin 8 open drain mode control, see P1.00OD	0
9	P1.09OD		Port 1 pin 9 open drain mode control, see P1.00OD	0
10	P1.10OD		Port 1 pin 10 open drain mode control, see P1.00OD	0
13:11	-		Reserved.	NA
14	P1.14OD		Port 1 pin 14 open drain mode control, see P1.00OD	0
15	P1.15OD		Port 1 pin 15 open drain mode control, see P1.00OD	0
16	P1.16OD ^[1]		Port 1 pin 16 open drain mode control, see P1.00OD	0
17	P1.17OD ^[1]		Port 1 pin 17 open drain mode control, see P1.00OD	0
18	P1.18OD		Port 1 pin 18 open drain mode control, see P1.00OD	0
19	P1.19OD		Port 1 pin 19 open drain mode control, see P1.00OD	0
20	P1.20OD		Port 1 pin 20 open drain mode control, see P1.00OD	0
21	P1.21OD ^[1]		Port 1 pin 21 open drain mode control, see P1.00OD	0
22	P1.22OD		Port 1 pin 22 open drain mode control, see P1.00OD	0
23	P1.23OD		Port 1 pin 23 open drain mode control, see P1.00OD	0
24	P1.24OD		Port 1 pin 24 open drain mode control, see P1.00OD	0
25	P1.25OD		Port 1 pin 25 open drain mode control, see P1.00OD	0
26	P1.26OD		Port 1 pin 26 open drain mode control, see P1.00OD	0
27	P1.27OD ^[1]		Port 1 pin 27 open drain mode control, see P1.00OD	0
28	P1.28OD		Port 1 pin 28 open drain mode control, see P1.00OD	0
29	P1.29OD		Port 1 pin 29 open drain mode control, see P1.00OD	0
30	P1.30OD		Port 1 pin 30 open drain mode control, see P1.00OD	0
31	P1.31OD		Port 1 pin 31 open drain mode control.	0

[1] Not available on 80-pin package.

8.5.18 Open Drain Pin Mode select register 2 (PINMODE_OD2 - 0x4002 C070)

This register controls the open drain mode for Port 2 pins. For details see [Section 8.4 "Pin mode select register values"](#).

Table 97. Open Drain Pin Mode select register 2 (PINMODE_OD2 - address 0x4002 C070) bit description

PINMODE_OD2	Symbol	Value	Description	Reset value
0	P2.00OD		Port 2 pin 0 open drain mode control.	0
		0	P2.0 pin is in the normal (not open drain) mode.	
		1	P2.0 pin is in the open drain mode.	
1	P2.01OD		Port 2 pin 1 open drain mode control, see P2.00OD	0
2	P2.02OD		Port 2 pin 2 open drain mode control, see P2.00OD	0
3	P2.03OD		Port 2 pin 3 open drain mode control, see P2.00OD	0
4	P2.04OD		Port 2 pin 4 open drain mode control, see P2.00OD	0

Table 97. Open Drain Pin Mode select register 2 (PINMODE_OD2 - address 0x4002 C070) bit description ...continued

PINMODE_OD2	Symbol	Value	Description	Reset value
5	P2.05OD		Port 2 pin 5 open drain mode control, see P2.00OD	0
6	P2.06OD		Port 2 pin 6 open drain mode control, see P2.00OD	0
7	P2.07OD		Port 2 pin 7 open drain mode control, see P2.00OD	0
8	P2.08OD		Port 2 pin 8 open drain mode control, see P2.00OD	0
9	P2.09OD		Port 2 pin 9 open drain mode control, see P2.00OD	0
10	P2.10OD		Port 2 pin 10 open drain mode control, see P2.00OD	0
11	P2.11OD ^[1]		Port 2 pin 11 open drain mode control, see P2.00OD	0
12	P2.12OD ^[1]		Port 2 pin 12 open drain mode control, see P2.00OD	0
13	P2.13OD ^[1]		Port 2 pin 13 open drain mode control, see P2.00OD	0
31:14	-		Reserved.	NA

[1] Not available on 80-pin package.

8.5.19 Open Drain Pin Mode select register 3 (PINMODE_OD3 - 0x4002 C074)

This register controls the open drain mode for Port 3 pins. For details see [Section 8.4 “Pin mode select register values”](#).

Table 98. Open Drain Pin Mode select register 3 (PINMODE_OD3 - address 0x4002 C074) bit description

PINMODE_OD3	Symbol	Value	Description	Reset value
24:0	-		Reserved.	NA
25	P3.25OD ^[1]		Port 3 pin 25 open drain mode control.	0
		0	P3.25 pin is in the normal (not open drain) mode.	
		1	P3.25 pin is in the open drain mode.	
26	P3.26OD ^[1]		Port 3 pin 26 open drain mode control, see P3.25OD	0
31:27	-		Reserved.	NA

[1] Not available on 80-pin package.

8.5.20 Open Drain Pin Mode select register 4 (PINMODE_OD4 - 0x4002 C078)

This register controls the open drain mode for Port 4 pins. For details see [Section 8.4 “Pin mode select register values”](#).

Table 99. Open Drain Pin Mode select register 4 (PINMODE_OD4 - address 0x4002 C078) bit description

PINMODE_OD4	Symbol	Value	Description	Reset value
27:0	-		Reserved.	NA

Table 99. Open Drain Pin Mode select register 4 (PINMODE_OD4 - address 0x4002 C078) bit description ...continued

PINMODE_OD4	Symbol	Value	Description	Reset value
28	P4.28OD		Port 4 pin 28 open drain mode control.	0
		0	P4.28 pin is in the normal (not open drain) mode.	
		1	P4.28 pin is in the open drain mode.	
29	P4.28OD		Port 4 pin 29 open drain mode control, see P4.28OD	0
31:30	-		Reserved.	NA

8.5.21 I²C Pin Configuration register (I2CPADCFG - 0x4002 C07C)

The I2CPADCFG register allows configuration of the I²C pins for the I2C0 interface only, in order to support various I²C-bus operating modes. For use in standard or Fast Mode I²C, the 4 bits in I2CPADCFG should be 0, the default value for this register. For Fast Mode Plus, the SDADRV0 and SCLDRV0 bits should be 1. For non-I²C use of these pins, it may be desirable to turn off I²C filtering and slew rate control by setting SDAI2C0 and SCLI2C0 to 1. See [Table 100](#) below.

Table 100. I²C Pin Configuration register (I2CPADCFG - address 0x4002 C07C) bit description

I2CPADCFG	Symbol	Value	Description	Reset value
0	SDADRV0		Drive mode control for the SDA0 pin, P0.27.	0
		0	The SDA0 pin is in the standard drive mode.	
		1	The SDA0 pin is in Fast Mode Plus drive mode.	
1	SDAI2C0		I ² C mode control for the SDA0 pin, P0.27.	0
		0	The SDA0 pin has I ² C glitch filtering and slew rate control enabled.	
		1	The SDA0 pin has I ² C glitch filtering and slew rate control disabled.	
2	SCLDRV0		Drive mode control for the SCL0 pin, P0.28.	0
		0	The SCL0 pin is in the standard drive mode.	
		1	The SCL0 pin is in Fast Mode Plus drive mode.	
3	SCLI2C0		I ² C mode control for the SCL0 pin, P0.28.	0
		0	The SCL0 pin has I ² C glitch filtering and slew rate control enabled.	
		1	The SCL0 pin has I ² C glitch filtering and slew rate control disabled.	
31:4	-		Reserved.	NA